Toward Better Out-painting : Improving the Image Composition with Initialization Policy Model

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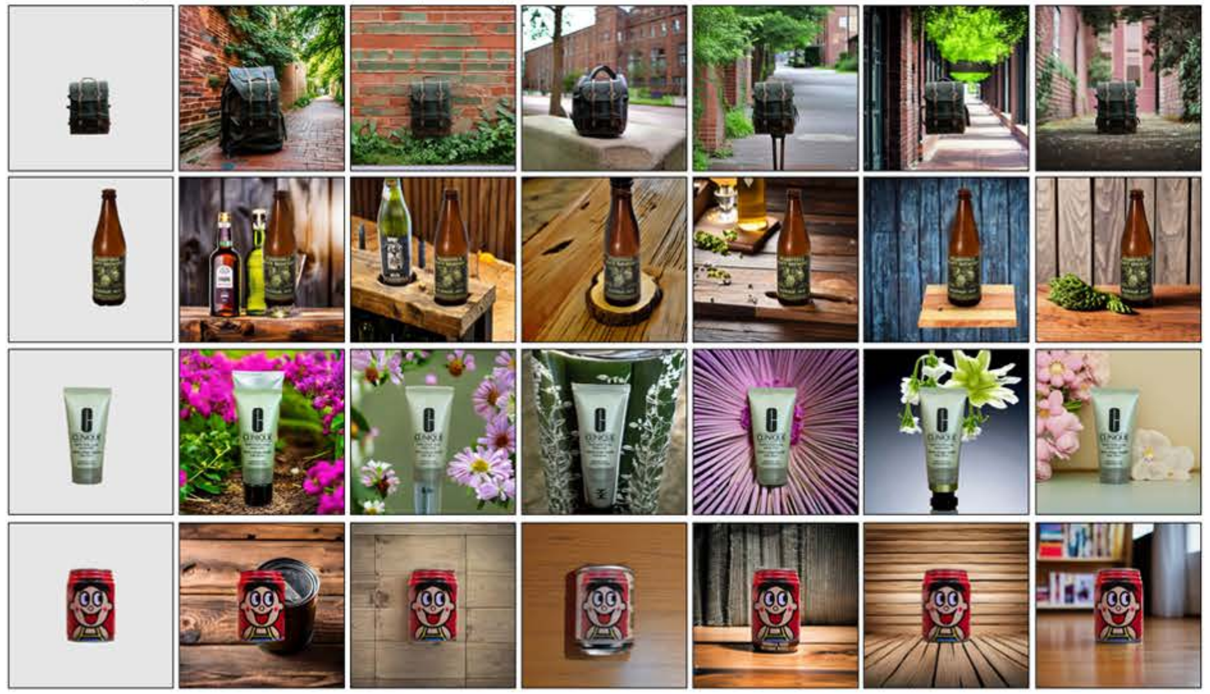
Comparison with Existing Methods



Variable Scene Description

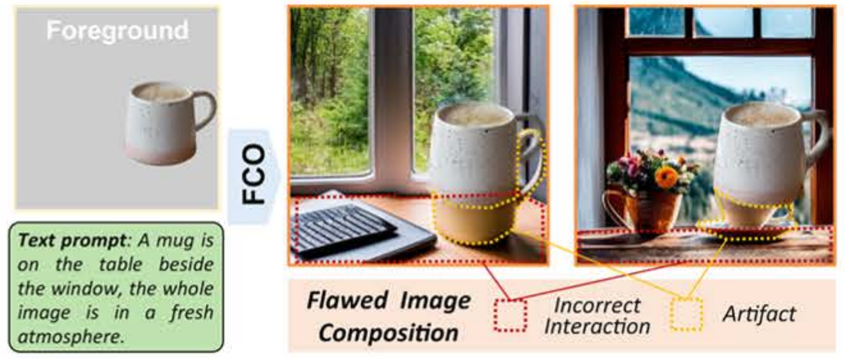
[*] on picnic blanket with canyon. [*] is on a windswept cliffside. [*] is placed with a potted plant. [*] sits on a rooftop patio. [*] stands on the bustling city street. [*] slices on a misty mountain road. [*] is on a tree-lined boulevard. [*] stands in the snowy scene.

Variable Object Pose, Position and Scale

- A Siamese cat stands on a stone, surrounded by purple flowers.in a sunlit garden.
- An apple sits atop a stack of books in a cozy library.
- A rabbit sits quietly on a windowsill during a rainy afternoon.
- A can sits on a wooden surface with a blurred ocean backdrop.

A. Motivation

A.1 The Flawed Image Composition in FCO Task

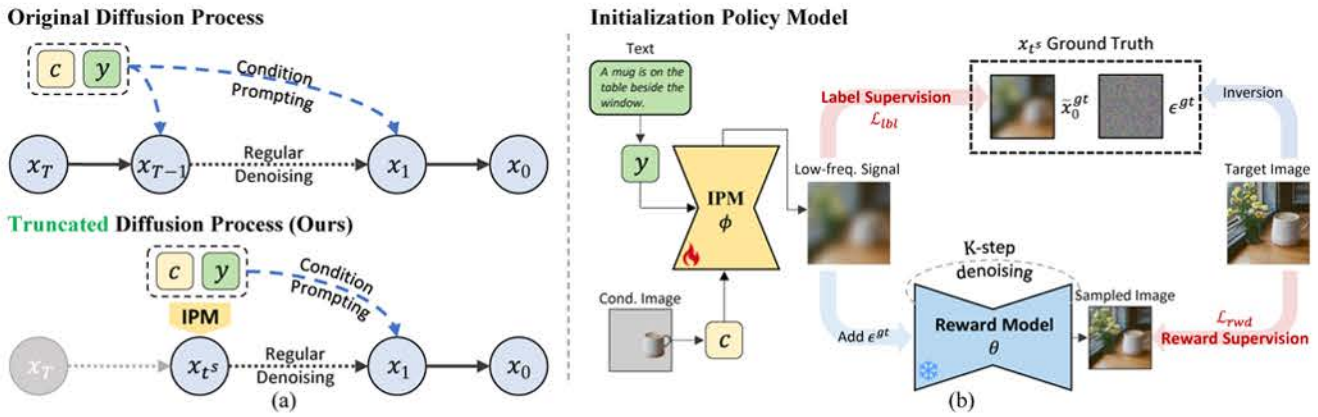


- The background objects like the table fail to interact appropriately with the given mug, alongside the artifacts generated in the process.

A.2 The Role of Initial Noise in This Problem

- In diffusion-based FCO, images are gradually denoised from initial noise x_T. The variations in initial noise x_T comprehensively influence the position, and pose of synthesized objects. The independently sampled x_T frequently implies compositions conflicting with the fixed foreground.

B. Methodology



B.1 Truncated Diffusion Model

- We propose truncating the denoising process at a specific time step t^s and substituting the early steps with an Initialization Policy Model. This method exploits the time-frequency coupling in the diffusion process, where the low-frequency signals that determine the image composition are mainly formed in the early stage.
- IPM is defined to be solely conditioned on the foreground image c and the text prompt y.

B.2 Initialization Policy Model (IPM)

- To facilitate the optimization of IPM, a comprehensive training paradigm integrating label supervision and reward supervision is proposed.
- Label Supervision uses inversion-derived low-frequency signal ground truth to guide IPM output.
- Reward Supervision subjects predicted x_{t^s} to multiple denoising rounds, with the error between the final image and target image serving as the reward criterion.

C. Implementation

C.1 Tricks and Implementation Details

- Network Initialization: The IPM φ is directly initialized from the U-Net of Stable Diffusion. This strategy aims to inherit the rich text-image priors embedded in its parameters.
- Ground Truth in Label Supervision: The ground truth in label supervision is derived through the DDIM Inversion.
- Fast Sampling in Reward Supervision: The diffusion reward adopts the denoising schedule of DDIM. The denoising steps K is set to 10.
- Stop Gradients in Reward Supervision: Due to multiple rounds of sampling operations having been performed, the gradient propagation of reward supervision requires repeated passage through the reward model θ. To save the video memory, we use the technique of stop gradients ($\partial x_{t-1} = \alpha_t \, \partial x_t$).
- Task-specific Dataset: We introduce a novel task-specific dataset built on OpenImage-v7, including 580K images with 800K masks, 6K unique object tags, and image captions.

C.2 The Effectiveness of Proposed Method

- A tube is placed in front of the delicate flowers, symbolizing freshness and purity.

- The generation results of the baseline model with or w/o IPM.

- The samples of predicted low-frequency signals and final generated images with or w/o the diffusion reward.

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